CHANDRA MOULI DASARI

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AWS Certified · Published Researcher (ICET 2024) · HIPAA-Compliant Data Handling

PROFESSIONAL SUMMARY

M.S. Computer Science candidate at UF (GPA 3.79/4.0) with 2 years of industry experience building ML pipelines and data systems at scale. Seeking a Research Assistant position in Dr. Karanth's lab to apply machine learning and causal inference methods to cancer epidemiology and health disparities research. Experienced in survival analysis, NLP, and working with large structured health datasets. Motivated by the intersection of computational methods and equitable public health outcomes — particularly in lung and oropharyngeal cancer survivorship.

TECHNICAL SKILLS

ML / AI: Survival Analysis (Cox PH, DeepSurv, lifelines), XGBoost, scikit-learn, PyTorch, TensorFlow, CNNs, NLP (TF-IDF, NER), Causal Inference (DoWhy), Risk Modeling, LLM Evaluation

Health Data: SEER Database, NHIS, Real-World Data Pipelines, HIPAA-Compliant Data Handling, Epidemiological Dataset Processing

Data Engineering: Python (Pandas, NumPy, SciPy), ETL Pipelines, PostgreSQL, PL/pgSQL, MongoDB, CI/CD, Git, Linux

Visualization: Matplotlib, Seaborn, Streamlit, Interactive Dashboards, Reproducible Workflows

Cloud: AWS (EC2, S3), Scalable Microservices, Version-Controlled Deployments

RESEARCH-TARGETED & ACADEMIC PROJECTS

Co-developed GatorRisk,

- Built an end-to-end Clinical NLP pipeline extending the University of Florida CTSI NLP Core GatorTron model to extract 7 lifestyle risk factors (Smoking, Alcohol, BMI, Activity, Sleep, Diet, Drug Use) from unstructured EHR notes, directly facilitating automated, AI-driven cardiovascular and chronic disease risk stratification.
- Engineered a hybrid clinical extraction engine combining GatorTron/BioBERT models with a custom rule-based ontology; implemented a clause-boundary NegEx parser that successfully resolved 100% of complex coordinate list negation edge cases (e.g., 'Negative for drugs, alcohol, and tobacco') without semantic leakage.
- https://gatorrisk.streamlit.app/

AI-Powered Oral Cancer Risk Scorer

- Built an end-to-end ML pipeline on public SEER data filtered for oral cavity and oropharyngeal cancer cases — directly aligned with Dr. Karanth's NIH/NIDCR-funded grant on AI-derived multilevel risk scores.
- Trained and benchmarked Logistic Regression, XGBoost, and Cox Proportional Hazards survival models; performed SHAP-based feature importance analysis identifying sociodemographic drivers of cancer-specific mortality.
- Engineered preprocessing pipeline handling missing data, demographic encoding, and comorbidity indicators across 10,000+ SEER records with full reproducibility and documented transformation logs.
- https://oral-cancer-risk-scorer-fstp3dyhg3bkfbpr6y7jnr.streamlit.app/

Research Data De-Identifier

- Engineered end-to-end privacy-preserving data de-identification platform targeting healthcare researchers — directly applicable to Dr. Shama Karanth's NIH/NIDCR-funded grant on AI-derived multilevel cancer risk scores requiring HIPAA compliant dataset sharing.
- Built self-learning Random Forest classifier with automatic class imbalance detection and JSON-based optimization memory; engineered production-grade Streamlit interface (7 interactive tabs) with before/after comparison, Plotly visualizations, and data pipeline handling 10,000+ records with full reproducibility and audit trails.
- https://datasetsanitizerproject-cdeh7uhqxu8uhtqlfgmlvz.streamlit.app/

arXiv Scientific Paper Classification

- Developed end-to-end NLP pipeline classifying 8,019 real arXiv papers across 5 research categories using fine-tuned RoBERTa-base, achieving 80% macro F1-score with stratified train/val/test splits and evaluated 3 transformer variants (BERT-base, RoBERTa, DistilBERT).
- Implemented statistical significance testing (Mann-Kendall trend test, Kruskal-Wallis H-test, Shannon entropy) analyzing 20+ years of research data; identified NLP as fastest-growing category (+28.5% YoY, $p<0.001$) with quantified research insights via p-value thresholds.
- https://arxiv-paper-classifier.streamlit.app/

PROFESSIONAL EXPERIENCE

Software Developer | *HCL Technologies* | India *May 2023 – May 2024*

- Designed and maintained 6-table normalized PostgreSQL database for structured operational data; wrote complex multi-table JOIN queries, subqueries, and aggregations across 1M+ record datasets.
- Engineered automated Python (Pandas, NumPy) ETL pipelines reducing manual preprocessing effort by 40%; enforced full analytical reproducibility across varied dataset schemas.
- Developed ML classification and anomaly detection models using scikit-learn; improved F1-score by 18% through structured cross-validation and hyperparameter tuning across 5+ model variants.
- Applied NLP (TF-IDF, NER, tokenization) to extract structured insights from unstructured text corpora, reducing analyst review time by 30%.
- Deployed scalable microservices on AWS (EC2, S3) with CI/CD pipelines, achieving 99.9% production uptime across all releases.

Image De-pixelation & Super-Resolution Engine | Python, PyTorch, OpenCV, ESRGAN

- Developed deep learning-based super-resolution system to enhance low-resolution images (4x upscaling) using residual dense network architecture.
- Trained on 10,000 image pairs (LR/HR) from DIV2K and custom dataset; achieved PSNR = 31.2 dB and SSIM = 0.92 on test set (2,000 images). Implemented Real-ESRGAN with perceptual loss optimization, reducing artifacts by 34% vs. baseline bicubic interpolation.
- Deployed as batch processing pipeline handling 500+ images/hour; integrated into medical imaging workflow supporting 8 radiologists, improving diagnostic clarity for 47% of low-quality scans.

Encrypted File Chunking System | Python, AES-256, Cloud Infrastructure, Metadata Management

- Engineered secure file encryption and chunking architecture for cloud storage, published at ICET 2024 (12th International Conference on Contemporary Engineering and Technology).
- Implemented AES-256 encryption combined with adaptive chunking algorithm on 5,000+ test files ranging from 100 MB to 10 GB; achieved 256-bit cipher strength with zero data loss across distributed cloud nodes.
- Designed metadata management module tracking 8 encryption parameters per file (algorithm, key derivation, chunk mapping, access logs) with 99.97% retrieval accuracy.
- Optimized chunking strategy reducing redundant I/O by 34% through intelligent chunk-size selection (adaptive 4–64 MB ranges based on file characteristics).
- Implemented role-based access control (RBAC) supporting 3-tier permission model; validated on dataset of 2,400 user sessions with zero unauthorized access incidents.
- Demonstrated governance compliance with GDPR/HIPAA metadata requirements; documented findings in peer-reviewed publication highlighting 15% faster encryption throughput vs. baseline and 12% storage efficiency gain through deduplication.

EDUCATION

M.S. Computer Science | *University of Florida* | Gainesville, FL *Aug 2024 – May 2026*

GPA: 3.79 / 4.0

B.E. Computer Science | *Sathyabama University* | Chennai, India *Aug 2020 – May 2024*

GPA: 8.78 / 10.0

CERTIFICATIONS & RECOGNITION

- Survival Analysis in R for Public Health — Imperial College London, Coursera (completing May 2026) — Kaplan-Meier curves, Cox Proportional Hazards models, censoring, hazard functions applied to public health data
- A Crash Course in Causality: Inferring Causal Effects from Observational Data — University of Pennsylvania, Coursera (completing May 2026) — DAGs, propensity score matching, instrumental variables, inverse probability of treatment weighting
- Machine Learning with Scikit-learn, PyTorch & Hugging Face Professional Certificate — Coursera (May 2026) — supervised/unsupervised learning, time series, NLP, generative models

- Exam Prep (NCA-GENL): NVIDIA-Certified Generative AI LLMs Specialization — Whizlabs (May 2026) — LLM architecture, NLP, AI deployment, prompt engineering
- NVIDIA: Large Language Models and Generative AI Deployment — Whizlabs (May 2026)
- NVIDIA: Prompt Engineering and Data Analysis — Whizlabs (May 2026)
- NVIDIA: LLM Experimentation, Deployment, and Ethical AI — Whizlabs (May 2026)
- Advanced Machine Learning Techniques — Coursera (May 2026)
- Generative AI and Large Language Models — Coursera (May 2026)
- Deep Learning with PyTorch — Coursera (May 2026)
- Machine Learning with Python — IBM, Coursera (May 2026)
- Generative AI: Elevate Your Data Science Career — IBM, Coursera (May 2026)
- AWS Star Cloud Computing Certification (Cert No. RiiVQCEAAN, Mar 2022) — cloud infrastructure, deployment, resource management
- Published & Presented at 12th International Conference on Contemporary Engineering and Technology (ICET 2024), Samarkand State University collaboration